

Design and Simulation of Wireless Charging System for Electrical Vehicles

B. Tech Final Year Project Work Presentation By

BANOTH NAVEEN (18JE0211)

J SAMRAT VARUN (18JE0371)

BHAWANI SHANKAR MEENA (18JE0225)

Under the Guidance of

Dr. Manodipan Sahoo

Assistant Professor



DEPARTMENT OF ELECTRONICS ENGINEERING, INDIAN INSTITUTE OF TECHNOLOGY (ISM)

Dhanbad-826004, India.

AGENDA

- ▶ **Objectives.**
- ▶ **Introduction.**
- ▶ **Overview of Electrical Vehicle.**
- ▶ **Overview of Wireless Charging System.**
- ▶ **Overview of Battery.**
- ▶ **Types of Wireless Charging System.**
- ▶ **Static and Dynamic Charging.**
- ▶ **Drawback of Wireless Charging System.**
- ▶ **Conclusion.**
- ▶ **Future Influence.**
- ▶ **Acknowledgement**

OBJECTIVES

- ▶ **To find the State of charge for the Wireless charging system.**
- ▶ **To give an overview of various wireless charging techniques out of which inductive wireless transfer has proven to be the best method of wireless charging.**
- ▶ **This also attempts to review the application of static and dynamic wireless charging and how the battery plays an important role in the electric vehicle.**

INTRODUCTION

What is the Wireless Power Transmission?

- ▶ The Transmission of Energy from one place to another without using wires.
- ▶ Conventional Energy Transfer is using wires.
- ▶ The Wireless Transmission is made possible by using various Technologies.
- ▶ Energy is Transmitted through electromagnetic waves over long distances.
- ▶ Like, wireless charging mobile phone, not on a grand scale.



Why is Wireless Power Transmission ?

- ▶ Technology will reduce the annoyance of wires.
- ▶ Improve the power transfer mechanisms.
- ▶ Recently, the wireless power transmission gain enormous attention to charging the on-board batteries of the Electric Vehicle(EV).
- ▶ It can transfer maximum power by achieving maximum efficiency.
- ▶ Wireless Power Transmission systems are classified as micro wave, evanescent wave, magnetic resonance, electrical resonance, or electromagnetic induction methods.

Why Electrical Vehicle?

- ▶ No Conventional Fuel.
- ▶ Cheaper to maintain.
- ▶ More eco-friendly, lower carbon footprint.
- ▶ Less noise pollution.
- ▶ Smoother Ride.



Benefits of Electric Vehicle

- ▶ Better Energy Efficient.
- ▶ Reduced Energy Dependence.
- ▶ Regenerative Braking.
- ▶ Easy to make autonomous vehicle.
- ▶ Easy Driving.
- ▶ Pollution Free.



Wireless Charging system

- ▶ Wireless charging uses an Electromagnetic Field to Transfer Energy between two objects through Electromagnetic Induction.
- ▶ Energy is sent through an inductive coupling to an Electrical device, which can then use that Energy to charge Batteries (or) run the device.



History of Wireless Charging



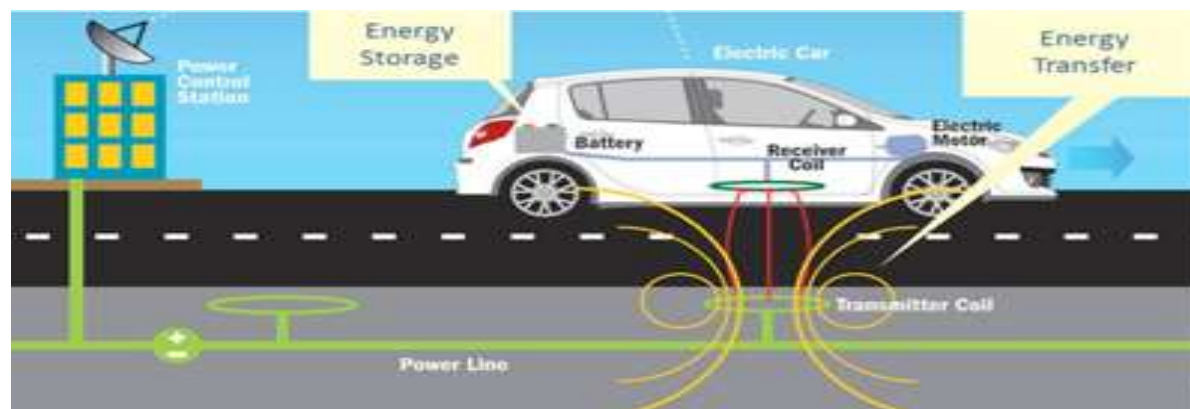
187 feet tall tower

History of wireless charging

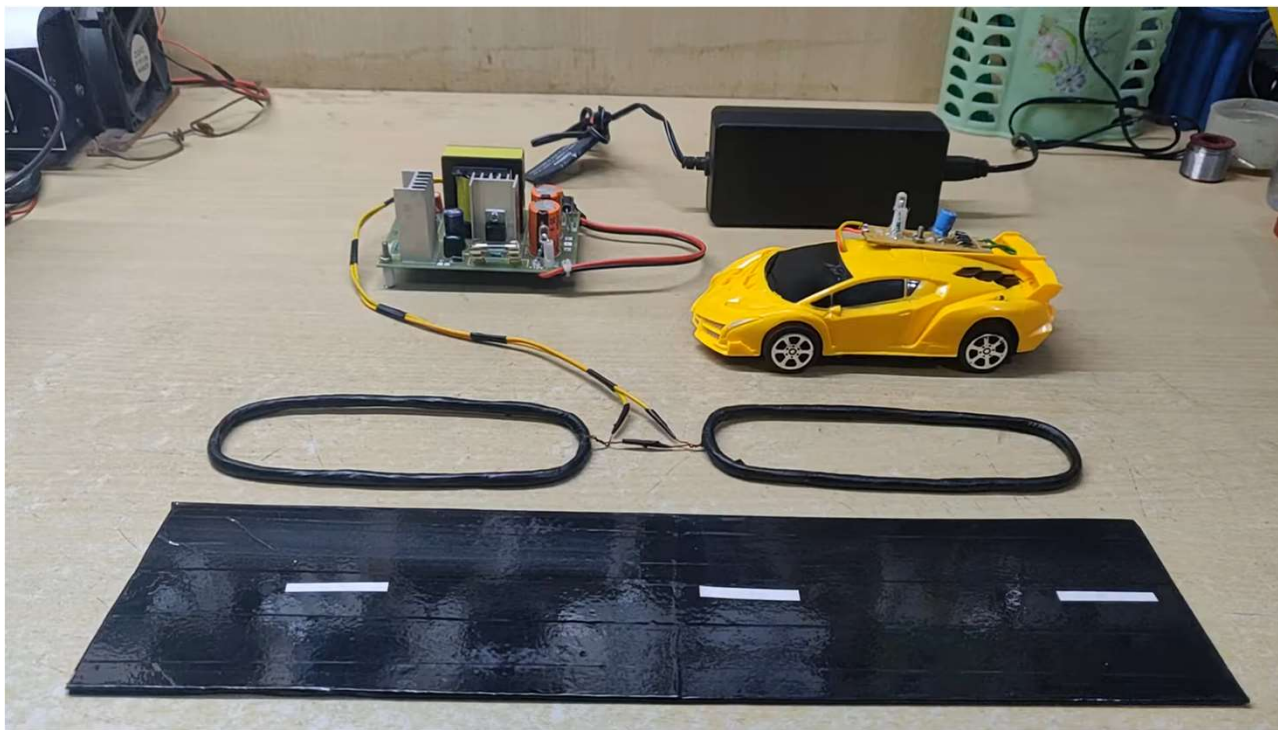
- ▶ In 1901, Nikola Tesla was able to transfer energy from one coil to another coil wirelessly.
- ▶ 187 feet tall tower to broadcast energy.
- ▶ He managed to light 200 lamps from a distance of 40 km.
- ▶ Due to a shortage of funds, the tower did not operate.
- ▶ The idea of Tesla is taken into research after 100 years by a team led by Marin Soljacic from MIT, The project is named “WiTricity”.

Working of Wireless Charging

- ▶ Specifically, the Technology works by Generating 20kHz frequencies into an Electromagnetic Field by a power Inverter that is Embedded.
- ▶ Wireless charging Electric Vehicles are built with an onboard pickup device that allows an electric vehicle to be powered by an underground powerline while driving.
- ▶ This allows the vehicle to carry 4/5 less Battery capacity or weight Because it only requires a small battery for minimal operations off the power grid.

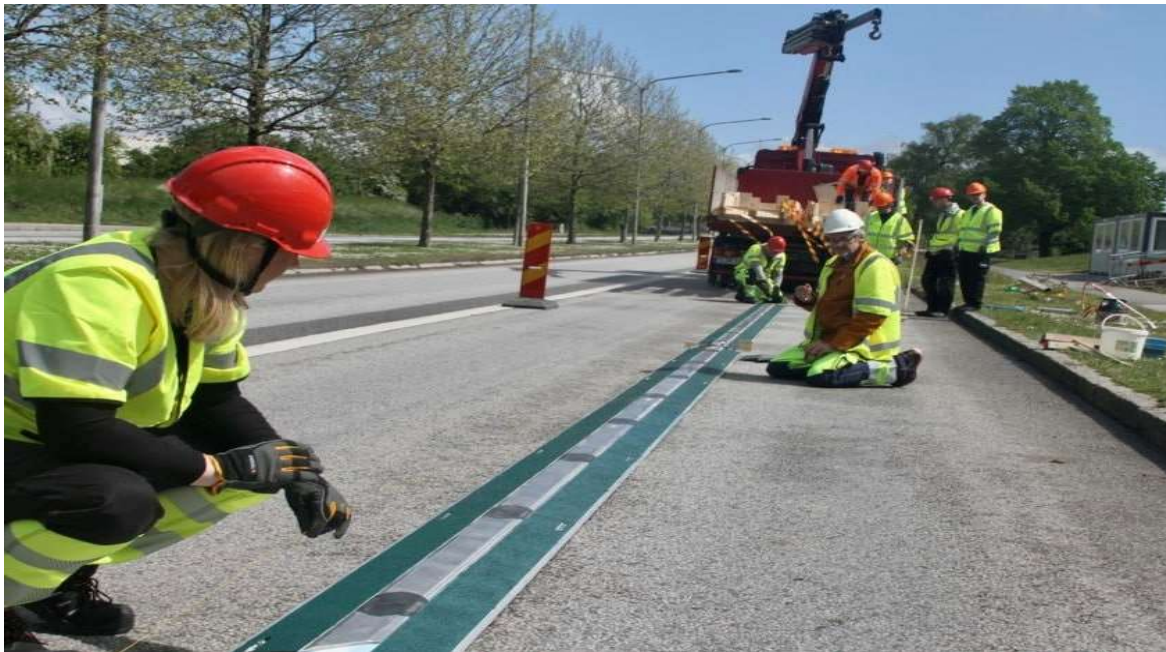


PROJECT WORKING DEMO



Construction of Roads for Electric Vehicle

- Electric Power Strips are buried 30cm Deep under the Road surface and Connected to the Power Grid.



Introduction to Battery

- ▶ Battery is the device which stores the energy in the form of chemical energy and gives electrical energy.
- ▶ Batteries have high energy density but low power density.
- ▶ For getting high energy number of cells are connected to form a battery pack.
- ▶ Example: Lead-acid battery, Li-ion battery.



Types of Battery

1) Non-Rechargeable(Primary)

- ▶ This type of cells can be used to get energy for one time.
- ▶ This cells works on spontaneous forward reaction.
- ▶ This product form can not be converted to the reactant by applying electric field.
- ▶ Examples: Dry cells, Alkaline cells, Li-Br battery.



2) Rechargeable(Secondary)

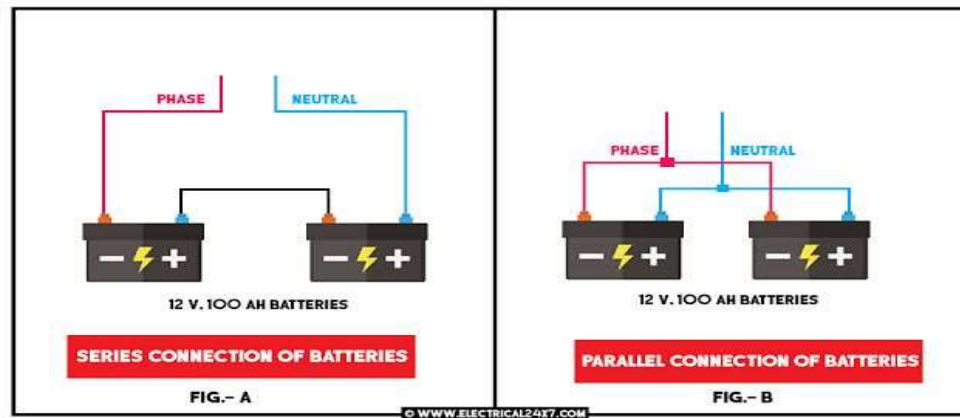
- ▶ This type of battery can be used again and again by charging it.
- ▶ This cells operates near the equilibrium and the reaction are reversible.
- ▶ Examples: Lead acid battery, Lithium ion battery.



Most commonly used battery

- ▶ For portable devices dry cells are used.
- ▶ The lead-acid batteries used in Vehicles.
- ▶ Lithium-ion batteries used for portable electronics such as laptops and smartphones.

Battery Banking



- ▶ For large amount of energy storage we need number of batteries.
- ▶ For this purpose batteries are connected in series and parallel.
- ▶ Series – voltage of battery bank is increased.
- ▶ Parallel - current increases.
- ▶ In battery bank, by using battery management system life can be increased.

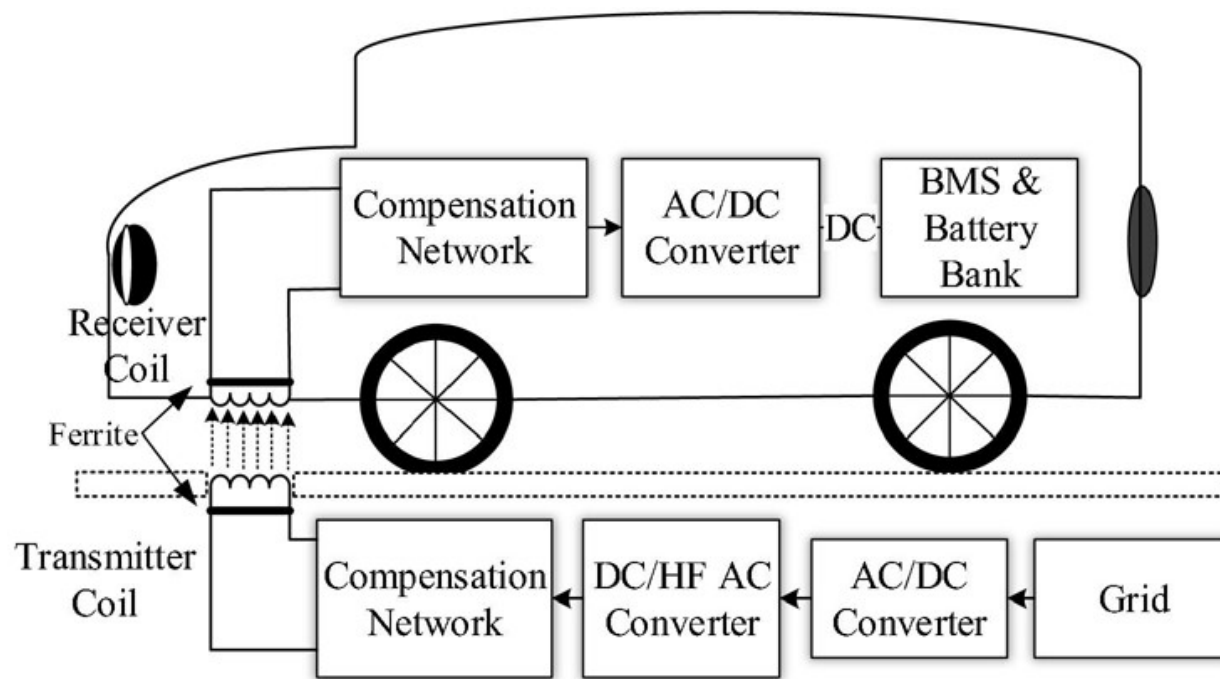
Advantages of Electrical Vehicle

- ▶ 80 % reduced operating cost than equivalent gas powered vehicles.
- ▶ Lower maintenance costs than gas powered vehicles.
- ▶ Pollution free.
- ▶ Zero recharging time and unlimited range(when operating on an grid road).
- ▶ Light weight Vehicles.
- ▶ A number of Electrical vehicle scan be charged at a time.
- ▶ Electrically safe.
- ▶ Charging is convenient.

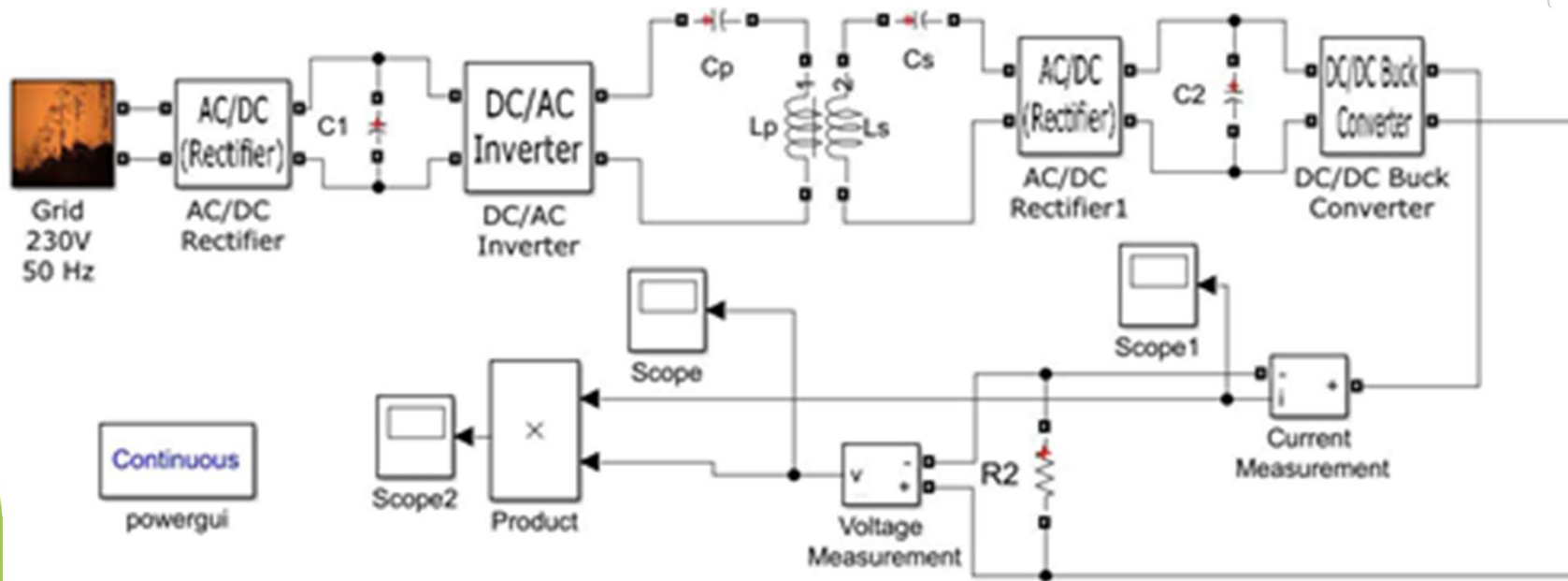
Disadvantages of Electrical Vehicle

- ▶ Initial Installation cost is very high.
- ▶ Working Area is limited.
- ▶ Currently the speed is limited to 40 miles/hour.
- ▶ Power outage might cause the Electrical vehicles to run out of charge.
- ▶ Heat generation is more than Traditional charging.
- ▶ Complex Design.
- ▶ Construction of wireless charging road is required.

Block Diagram of Wireless Charging System



Design of the Wireless Charging System in MATLAB



TYPES OF WIRELESS CHARGING SYSTEMS

- ▶ Based on operating techniques wireless charging of electric vehicles can be classified into four types:
- ▶ Capacitive Wireless power transfer (CWPT)
- ▶ Permanent Magnetic Gear Wireless power transfer (PMWPT)
- ▶ Inductive Wireless power transfer (IWPT)
- ▶ Resonant Inductive Wireless power transfer (RIWPT)

Capacitive Wireless Charging System (CWCS)

- ▶ Wireless transfer of energy between transmitter and receiver is accomplished by means of displacement current caused by the variation of the electric field. Instead of magnets or coils as transmitters and receivers, coupling capacitors are used here for wireless transmission of power. The AC voltage was first supplied to the power factor correction circuit to improve efficiency and maintain the voltage levels and to reduce the losses while transmitting the power. Then it is supplied to an H-bridge for the High-frequency AC voltage generation and this high-frequency AC is applied to the transmitting plate which causes the development of the oscillating electric field that causes displacement current at the receiver plate by means of electro static induction.
- ▶ The AC Voltage at the receiver side is converted to DC to feed the battery through BMS by rectifier and filter circuits. Frequency, voltage, size of coupling capacitors, and air gap between transmitter and receiver affect the amount of power transferred. Its operating frequency is between 100 to 600 kHz.

Permanent Magnet Gear Wireless Charging System (PMWC)

- ▶ Here transmitter and receiver each consist of armature winding and synchronized permanent magnets inside the winding. At the transmitter, side operation is similar to motor operation. When we apply the AC current to transmitter winding it induces mechanical torque on the transmitter magnet causing its rotation. Due to the magnetic interaction change in the transmitter, the PM field causes torque on receiver PM which results from its rotation being synchronous with the transmitter magnet.
- ▶ Now change in receiver permanent magnetic field causes the AC current production in winding i.e, receiver acts as a generator as mechanical power input to the receiver PM converted into electrical output at receiver winding. The coupling of rotating permanent magnets is referred to as **magnetic gear**. The generated AC power at the receiver side is fed to the battery after rectifying and filtering through power converters.

Inductive Wireless Charging System (IWC)

- ▶ The basic principle of IWC is Faraday's law of induction. Here wireless transmission of power is achieved by mutual induction of magnetic field between transmitter and receiver coil. When the main AC supply is applied to the transmitter coil, it creates AC magnetic field that passes through the receiver coil and this magnetic field moves electrons in the receiver coil causing AC power output.
- ▶ This AC output is rectified and filtered to Charge the EV's energy storage system. The amount of power transferred depends on frequency, mutual inductance, and distance between transmitter and receiver coil. The operating frequency of IWC is between 19 to 50 kHz.

Resonant Inductive Wireless power transfer (RIWPT)

- ▶ Basically resonators having high-Quality factors transmit energy at a much higher rate, so by operating at resonance, even with weaker magnetic fields we can transmit the same amount of power as in IWC. The power can be transferred over long distances without wires.
- ▶ Max transfer of power over the air happens when the transmitter and receiver coils are tuned i.e., both coil's resonant frequencies should be matched. So to get good resonant frequencies, additional compensation networks in the series and parallel combinations are added to the transmitter and receiver coils.
- ▶ This additional compensation network along with improvement in resonant frequency also reduces the additional losses. The operating frequency of RIWC is between 10 to 150 kHz.

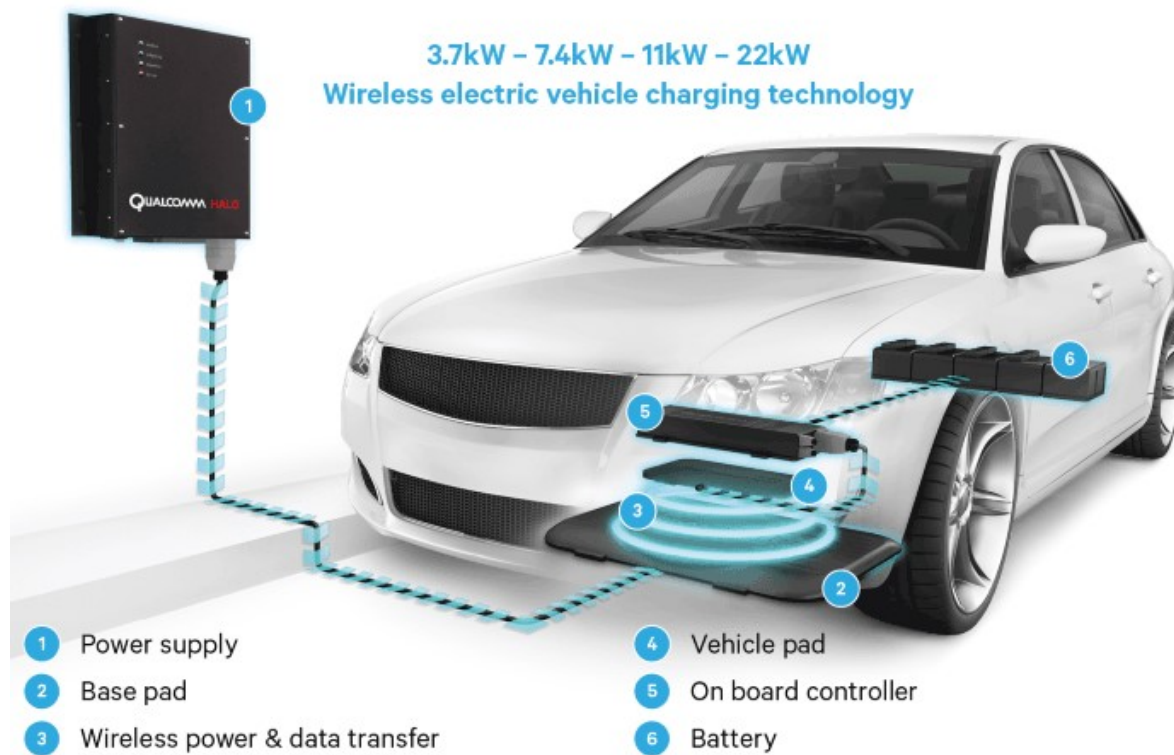
Static and Dynamic Wireless Charging

- ▶ Based on the application, Wireless charging systems for EV can be distinguished into two categories,
 1. Static Wireless Charging.
 2. Dynamic Wireless Charging.

1. Static Wireless Charging

- ▶ As the name indicates, the vehicle gets charged when it remains static. So here we could simply park the EV at the parking spot or in the garage which is incorporated with WCS. The transmitter is fitted underneath the ground and the receiver is arranged in vehicles underneath. To charge the vehicle align the transmitter and receiver and leave it for charging. The charging time depends on the AC supply power level, the distance between the transmitter & receiver, and their pad sizes.

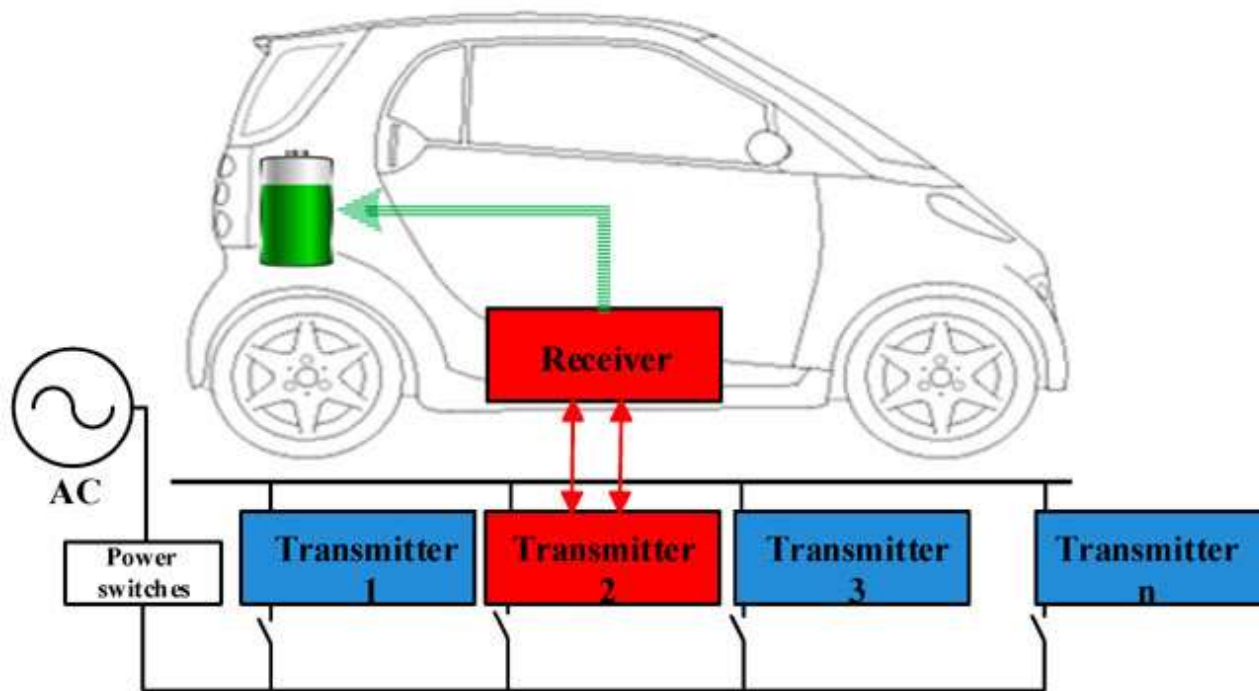
Static Wireless Charging



Dynamic Wireless Charging

- ▶ As the name indicates here the vehicle gets charged while in motion. The power transfers over the air from a stationary transmitter to the receiver coil in a moving vehicle. By using DWCS EV's traveling range could be improved with the continuous charging of its battery while driving on roadways and highways. It reduces the need for large energy storage which further reduces the weight of the vehicle.

Dynamic Wireless Charging



DRAWBACKS OF WIRELESS ELECTRIC VEHICLE CHARGING SYSTEM (WEVCS)

- ▶ **1. Inefficient transfer of electricity:**
- ▶ The energy efficiency via wired charging of the electric vehicles is somewhere between 90 – 93%, which makes it obvious that efficiency from wireless charging of EVs will be much less. However, with time the efficiency is likely to be improved.

- ▶ **Table -1** International Society of Automotive Engineers standards for wireless charging of electric vehicles

Wireless Power Transfer Levels	WPT-1	WPT-2	WPT-3	WPT-4
Maximum Input Power	3.7	7.7	11	22
Minimum Target Efficiency	>84%aligned			
Operating Frequency	85 (Band: 81.39-90)			

DRAWBACKS OF WIRELESS ELECTRIC VEHICLE CHARGING SYSTEM (WEVCS)

► **2. Health and safety issues:**

- As wireless charging of electric vehicles works at high voltage and current levels, safety concerns become most important. We definitely need strict monitoring in order to transfer so much electricity wirelessly without having adverse effects on living organisms. Besides that, the long-term effect of the electromagnetic field can also affect the living organisms which need to be inspected thoroughly. However, it also comes with three important potential health and safety issues namely electrical, magnetic, and fire hazards. Sometimes due to breakdown of devices or physical damage or environmental conditions, it can also produce electrical shocks.
- While working at higher power levels, it is possible for magnetic flux to exceed the minimum value given by the standard authority. In such cases, flora and fauna become vulnerable and are exposed to probable danger. Since there is a high-power transfer from the transmitter to the receiver charging pads at a substantially big air gap of about 150 to 300mm at some kilohertz to some Megahertz. As a result, high-frequency leakage fluxes are generated due to air gaps. The level of such exposure fluxes must be below or just follow the guidelines set by standard organizations.

Characteristics of various wireless power transfer methods

WirelessPower Transfer Methods	Capacitive	PermanentMagnetic Gear	Inductive	ResonanceInductive
Efficiency	Low/ Medium	Low/ Medium	Medium/ High	Medium/High
EMI	Medium	High	Medium	Low
Frequencyrange (KHz)	100-600	0.05-0.50	10-50	10-150
Size volume	Low	High	Medium	Medium
ComplexityOf design	Medium	High	Medium	Medium
Powerlevels	Low	Medium/ low	Medium/ High	Medium/low
Suitabilityfor WEVCS	Low/ Medium	Low/ Medium	High	High

CONCLUSION

- ▶ Wireless charging of electric vehicles has the potential to revolutionize road transportation in the automotive industry. With the advancement of electric vehicle technology, the wireless charging technique is expected to increase significantly by the next decade.
- ▶ The main agenda of this is to give an overview of various wireless charging techniques out of which inductive wireless transfer has proven to be the best method of wireless charging.
- ▶ This also attempts to review the application of static and dynamic wireless charging and how the battery plays an important role in the electric vehicle.
- ▶ Here, the battery size is affected by wireless charging techniques which lower the overall cost of the electric vehicle.
- ▶ The electric vehicle batteries which were to take quite a lot of time to charge up to the rated value will be charged within less time comparatively as their battery capacity is reduced.
- ▶ However, simplicity and minimum driver intervention are key features that win out time and time again, and when these features are coupled with high power transfer efficiency, wireless charging of electric vehicles is a winning combination.

FUTURE INFLUENCE OF WIRELESS CHARGING OF ELECTRIC VEHICLE

- ▶ Wireless Charging of Electric Vehicle, approach revolutionizes the changes in the electric vehicle Industry. In the last few years, trends suggest a rise in interest among the common masses for electric cars in comparison to electric two-wheelers and internal combustion engine vehicles or petrol/diesel cars.
- ▶ At a fundamental level, electric cars comparatively offer a lower operating cost compared to conventional internal combustion engines. On average, electric vehicles are 75- 80% cheaper from a fuel and maintenance perspective, which is an important factor for many consumers who have high usage.
- ▶ This reality holds true across factors because it's materially cheaper to charge a battery compared to refueling a conventional liquid fuel tank. The electric vehicles comparatively induced with dynamic wireless charging technique takes lesser time than the electric vehicles which are charged statically. With reduced new battery capacity using a dynamic wireless charging system electric vehicles can be charged under motion.

FUTURE INFLUENCE OF WIRELESS CHARGING OF ELECTRIC VEHICLE

- ▶ Comparatively, wireless charging system range and cost are the setbacks plug-in charging systems are facing currently. With the growing impact of fuel-driven vehicles on the environment, electric taxi fleets in congested urban streets are critical in cutting transport emissions and cleaning up the air.
- ▶ With reduced new battery capacity using a dynamic wireless charging system electric vehicles can be charged under motion.
- ▶ With more people opting for electric mobility, the new technology can also be rolled out broadly for public use. This in turn would help everyday drivers of electric cars charge more conveniently on the go. However, the time taken to charge wirelessly could reduce a taxi driver's earning potential, a problem that is known and being worked upon.

Wireless Electric Vehicle Charging Standards

- ▶ Wireless charging makes EV charge without any need for a plug-in. If every company makes its own standards for wireless charging systems that won't be compatible with other systems then it won't be a good thing. So to make wireless EV charging more user-friendly.
- ▶ Many international organizations like the International Electro-Technical Commission (IEC), the Society of Automotive Engineers (SAE), Underwriters Laboratories (UL) Institute of Electrical and Electronics Engineers (IEEE) are working on standards.

Charging Standards

- ▶ SAE J2954 defines WPT for Light-Duty Plug-In EVs and Alignment Methodology. According to this standard, level 1 offers maximum input power of 3.7 kW, level2 offers 7.7Kw, level 3 offers 11Kw and level4 offers 22Kw. And the minimum target efficiency must be greater than 85% when aligned. Allowable ground clearance should be up to 10 inches and side-to-side tolerance is up to 4 inches. The most preferable alignment method is magnetic triangulation which assists to stay within charge range in manual parking and assists to find parking spots for autonomous vehicles.
- ▶ SAE J1772 standard defines EV/PHEV Conductive Charge Coupler.
- ▶ SAE J2847/6 standard defines Communication Between Wireless Charged Vehicles and Wireless EV Chargers.
- ▶ SAE J1773 standard defines EV Inductively Coupled Charging.
- ▶ SAE J2836/6 standard defines Use Cases for Wireless Charging Communication for PEV.
- ▶ UL subject 2750 defines Outline of Investigation, for WEVCS.
- ▶ IEC 61980-1 Cor.1 Ed.1.0 defines EV WPT Systems General Requirements.
- ▶ IEC 62827-2 Ed.1.0 defines WPT-Management: Multiple Device Control Management.
- ▶ IEC 63028 Ed.1.0 defines WPT-Air Fuel Alliance Resonant Baseline System Specification.

Companies Currently Developed and Working on WCS

- ▶ **Evatran group's** making plug-less Charging for passenger EVs like Tesla Model S, BMW i3, Nissan Leaf, Gen 1 Chevrolet Volt.
- ▶ **WiTricity Corporation** has been making WCS for Passenger cars and SUVs till now it is working with Honda Motor Co. Ltd, Nissan, GM, Hyundai, and Furukawa Electric.
- ▶ **Qualcomm Halo** is making WCS for Passenger, sports, and race cars and it is acquired by WiTricity corporation.
- ▶ **Hevo Power** is making WCS for Passenger cars.
- ▶ **Bombardier Pre-move** is making WCS for Passenger cars to SUVs.
- ▶ **Siemens and BMW** are making WCS for Passenger cars.
- ▶ **Momentum Dynamic** is making WCS Corporation Commercial fleet and Bus.
- ▶ **Conductix - Wampfler** is making WCS for Industry fleet and Bus.

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